Alert-Budgeted Clinical Early Warning: Patient-Blocked Risk Calibration for Stateful Sepsis Alarm Policies

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Abstract

Clinical early warning must balance detection against a manageable alert load, yet threshold-independent discrimination metrics do not specify how often a deployed rule will alarm. We formulate early warning as a stateful alarm policy with a threshold, refractory window and escalation tier, and calibrate it against a patient-average budget on alarms per day. The artificial-intelligence contribution is a decision layer that converts any fixed risk score into a calibrated, auditable alarm policy: patient-level losses are combined with a finite-grid upper-confidence bound, a Learn-then-Test multiple-testing correction, and an optional randomised mixture of adjacent thresholds that can use a discrete grid's budget more fully. The engineering application is a causal sepsis-onset pipeline evaluated on 40,336 PhysioNet 2019 records from two hospitals, using prefix logistic, LightGBM and masked GRU scorers over three subject-disjoint seeds. On site A, mean sensitivity rises from 0.185 at 0.25 alarms per patient-day to 0.774 at 2 per day. Applying the source threshold to site B without target calibration lowers sensitivity at 1 per day from 0.585 to 0.315, even though the realised rate there is lower. Repeated-split coverage, label-free rate matching and a ward-load simulation quantify the price of the guarantee: the certificate is conservative at tight budgets and coarse grids, and target labels or local rate adaptation are needed to recover the operating point. The results support patient-blocked budget calibration as an auditable engineering instrument, but they do not establish clinical effectiveness, alert-fatigue reduction or deployment readiness.

Keywords: Clinical early warning, Sepsis onset, Alarm budgets, Stateful policies, Risk calibration, Clinical time series

1. Introduction

Deterioration events in hospitalised patients unfold over hours, and the physiological trajectory that precedes them is recorded long before the event itself is recognised. This temporal structure has motivated a substantial literature on data-driven early warning, ranging from static severity scores to sequential models that forecast deterioration trajectories from electronic health records [1] and compact, efficiency-aware networks that predict sepsis onset from structured time series [2]. Reported discrimination is high and improving, with in-hospital mortality and onset prediction now routinely exceeding the performance of classical scoring instruments [3]. The implicit promise of this line of work is that a sufficiently accurate model, once installed at the bedside, will shorten time to intervention.

Discriminative prediction and operational alerting address different questions. A score-to-alert conversion requires a decision rule, whereas threshold-independent ranking metrics do not specify how frequently that rule emits alarms. Reviews of intensive-care adverse-event prediction discuss incomplete reporting of alert burden, time-to-alert and deployment calibration [4]. Reviews of multivariate clinical time-series learning also motivate evaluation beyond architectural comparisons [5, 6]. Alert-fatigue studies reach the same conclusion from the workflow side: an alert that is statistically accurate but too frequent or too poorly targeted is dismissed, ignored or actively suppressed [7, 8], and intensive-care monitors can emit hundreds of alarms per patient-day, most of them clinically irrelevant [9]. These concerns support reporting the alarm policy and its realised load explicitly, rather than treating an area-under-the-curve comparison as a specification of bedside alerting behaviour.

Recent research improves the backbone that produces the risk score. For irregularly sampled and heterogeneously missing observations, approaches include reliability-aware temporal aggregation [10, 11], graph formulations over irregular observations [12], and masked-attention models for deterioration prediction [13]. Hybrid sequence-graph

models report strong discrimination on the 2019 cohort [14], structure-preserving models attempt to keep each alert connected to its evidence [15], and operational early-warning indices are being developed outside the public benchmark [16]. Transfer-oriented variants align feature spaces across cohorts [17], while dynamic temporal graphs incorporate clinical calculators into the prediction pathway [18]. These methods motivate prospective backbone comparisons, but their adapters are not included in the current reference implementation. Our implemented contribution instead concerns the causal mapping from fixed scores to stateful alarms and the risk associated with selecting that mapping.

A separate literature attaches reliability estimates to clinical predictions. Selective prediction and calibration have been analysed for multimodal clinical classification [19], reliability estimation has been extended beyond output-space calibration for time-series classifiers [20], and conformal methods have been used for physiological regression uncertainty [21]. Conformal risk control extends coverage to a user-specified expected loss [22], and recent work relaxes its monotonicity requirements and studies cross-validation variants [23, 24]. Time-series conformal methods also model temporal dependence directly [25, 26]. Our distinction is the unit and estimand of alarm-policy calibration: refractory state makes successive emitted alarms dependent, so those hours cannot simply be counted as independent patient samples. This does not imply that every conformal treatment of time series is invalid or that a stateful policy cannot be calibrated at the trajectory level. The reference construction uses independent patient blocks and fixed finite-grid upper-confidence bounds, a stronger sampling assumption than exchangeability alone.

This paper makes the alarm budget an explicit constraint and calibrates the policy that generates alerts. The implementation separates causal score estimation from full-trajectory policy replay. Its finite-sample bound concerns the population mean of patient-level rates, not a hard maximum for each patient or ward. We evaluate the procedure on the public 2019 A/B cohort with three subject-disjoint seeds, including a same-threshold external audit, a deliberately invalid timestep comparator and a target label-budget analysis. The contributions are as follows.

- We formulate early warning as a stateful alarm policy in which the refractory window and escalation tier are control variables, so that alarm volume becomes a design parameter of the monitoring system rather than an emergent property of a threshold.
- We derive a patient-blocked finite-grid upper-confidence bound on the population alarm rate under independent sampling, and compare it with a Learn-then-Test multiple-testing selector, a randomised adjacent-threshold mixture and a deliberately invalid timestep effective-sample-size shortcut.
- We provide causal reference learners, official-format data readers and cached-score experiment drivers, and report a real-data evaluation under shared requested budgets, including a cross-hospital transport audit and an explicit negative result for architecture-only claims.
- We quantify the target label budget required for threshold recalibration and add a label-free rate-matching rule: on this cohort, about 10% of eligible target subjects recover the full-pool operating point at 1 alarm per patient-day, while the label-free rule recovers the target operating point at moderate budgets without any target labels.
- We benchmark LightGBM and partial NEWS, SIRS and qSOFA under the same patient-level alarm budget, showing that a stronger tabular scorer improves external sensitivity without removing the transport gap, and that coarse integer instruments expose a threshold-grid failure mode.
- We report repeated-split coverage, PPV/NNA, a ward-load simulation and one-at-a-time ablations, so that the certificate's conservatism and the operational meaning of a budget are measured rather than asserted.

The remainder of the paper reviews related work (Section 2), develops the policy formulation and calibration procedure (Section 3), specifies the datasets and experimental protocol (Section 4), reports the results (Section 5), and discusses their scope and limitations (Section 6).

2. Related Work

2.1. Backbones for Irregularly Sampled Clinical Time Series

Recent work models irregular sampling explicitly, using continuous-time formulations [11], reliability-aware temporal aggregation [10], and graphs over observations and variables [12]. Hybrid sequence-graph designs report strong

discrimination on the 2019 cohort [14], while structure-preserving sequence models attempt to keep each alert connected to its supporting evidence [15]. Complementary approaches model the observation process to synthesise irregular trajectories [27] or exploit informative missingness for ward deterioration prediction [13]. Domain-specific applications include venous thromboembolism detection after cancer treatment [28], while surveys synthesise the wider modelling landscape [29]. These methods are potential future score backbones, not models already integrated or reproduced in the current package. Any integration must retain prefix-only information and cannot reuse a full-trajectory imputation procedure without a separate causality audit.

2.2. Early Warning and Deterioration Prediction

Clinical early warning has moved from static severity scores toward dynamic, trajectory-aware prediction. Classical early-warning scores remain important baselines and have been externally validated in emergency and prehospital settings [30, 31], but they are typically fixed-weight instruments rather than calibrated policies. Recent multicentre work predicts the shape of sepsis deterioration rather than a single risk estimate, arguing that heterogeneous trajectories require patient-specific forecasting [1], while efficiency-oriented designs pursue compact models that remain explainable under the compute constraints of a ward workstation [2]. Others inject clinical reasoning into the prediction pathway: dynamic temporal graphs embed calculator-derived features so that the model inherits the semantics of established instruments [18], and language-model-guided simulation is used to generate interpretable physiological dynamics for early warning [32]. Transfer-oriented formulations address the practical reality that a model trained in one institution must be reused in another with different coding practice and case mix [17], and earlier work established that deep models over longitudinal inpatient records can outperform tabular baselines for outcome prediction [3, 33, 34]. Prospective implementation studies show that deployment changes the evidentiary question: a model that performs well offline may produce modest clinical effects and a substantial alert burden in practice [35]. Multisite and external validations of sepsis models likewise report performance degradation and calibration drift across hospitals [36, 37]. Across this literature the model is usually the unit of contribution; the alerting rule that converts its output into an operational signal is treated as an afterthought, and comparison between methods is almost always conducted at a fixed threshold rather than at a fixed alarm volume.

2.3. Calibration, Selective Prediction, and Conformal Guarantees

Calibration and selective prediction have been studied in multimodal clinical classification [19]. Reliability estimation for time-series classifiers also examines evidence beyond output remapping [20], while adaptation audits consider realistic label delays [38]. Conformal risk control provides distribution-free control of an expected loss and has been extended to non-monotone losses and cross-validation variants [22–24]. Learn-then-Test makes the multiple-testing structure explicit: candidate configurations are null hypotheses, and a valid p-value turns rejection into a finite-sample risk certificate [39]; anytime-valid and quantile variants extend the same logic to optional stopping and tail risks [40, 41]. Time-series conformal methods handle temporal dependence explicitly [25, 26, 42], and prediction-set design has been studied for efficiency and human decision quality rather than coverage alone [43, 44]. Conformal methods have also been applied to physiological regression uncertainty [21], to performance estimation under distribution shift [45], and to patient-specific haemodynamic adaptation [46]. These contributions concern different units, targets and sampling assumptions; their guarantees cannot be declared invalid merely because an unrelated downstream alarm rule has memory. The present reference procedure instead specifies a patient-average alarm-rate estimand and uses independent-block concentration, without claiming the weaker assumptions or different risk statement of standard conformal constructions.

2.4. Evaluation Practice

Risk-of-bias instruments for prediction models address artificial-intelligence-specific issues in calibration, heterogeneity and applicability [47]. Reviews of intensive-care adverse-event and time-series prediction also discuss reporting and evaluation limitations [4–6]. The alert-fatigue literature adds a complementary measurement problem: dismissal, override and workflow burden are not captured by discrimination or calibration [7, 8, 48]. Industrial alarm management treats load, prioritisation and rationalisation as lifecycle metrics [49, 50]; clinical early warning has analogous goals but no shared patient-day rate budget. Data-sharing histories and foundation-model agendas also shape which cohorts are available for replication [51], and task heterogeneity across diagnoses complicates direct transfer

of mortality models [52, 53]. These concerns motivate a protocol that reports requested budgets, realised alarm rates, sensitivity and lead time alongside discrimination. The present evaluation follows that protocol on the public 2019 cohort; shared requested budgets make the operational constraint explicit but do not imply identical realised alert volume across models.

3. Method

3.1. Problem Setting and Notation

Let patient i contribute observations $O_i = \{(t_{ij}, m_{ij}, v_{ij})\}_{j=1}^{n_i}$, with observation times t_{ij}, variable indices m_{ij}, and values v_{ij}. The number and spacing of observations may differ across patients and variables. Decisions are made on a prespecified regular grid with sampling interval Δ hours, using only observations available at each decision time. A causal score function maps this history to $s_{it} = f_\theta(O_{i,\le t}) \in [0, 1]$; any preprocessing, including imputation, must satisfy the same restriction.

An observed event has time $t_i^\star$, and a useful early alarm falls in $[t_i^\star - W, t_i^\star)$, where $W = 12$ hours is a design choice for the proposed evaluation. Records without an observed event are represented separately from records whose event time is outside observed follow-up. Clinical event time and the boundary of a shifted prediction label are not interchangeable. The data protocol in Section 4 therefore requires an explicit onset convention and does not evaluate lead time against an unobserved or left-censored clinical onset.

3.2. Stateful Alarm Policies

The policy $\pi_\lambda = (\lambda, \tau, \kappa)$ comprises a threshold, an integer refractory duration, and an integer escalation duration. Here $\tau \ge 0$ is the number of subsequent decision steps suppressed after an alarm; $\kappa = 0$ disables escalation, while $\kappa > 0$ permits an override after κ consecutive above-threshold steps following the last alarm. Let z_{it} be the number of refractory steps remaining at the start of step t, and let c_{it} include the current exceedance and the consecutive suppressed exceedances immediately preceding it. With initial state zero, the emission and update are

$$\begin{aligned} a_{it} &= \mathbf{1}[s_{it} \text{ finite},\ s_{it} \ge \lambda,\ z_{it} = 0 \ \vee\ (\kappa > 0 \ \wedge\ c_{it} \ge \kappa)]\,, \\ z_{i,t+1} &= \begin{cases} \tau, & a_{it} = 1, \\ \max(0, z_{it} - 1), & a_{it} = 0. \end{cases} \end{aligned} \tag{1}$$

The exceedance counter resets after an alarm or a non-exceedance, including a missing score. The refractory duration counts decision steps, not observations in the irregular raw record. Setting $\tau = 0$ gives the memoryless threshold policy, whereas $\lambda = +\infty$ gives an exact no-alarm policy. The latter is an explicit safe fallback and is distinct from the largest finite grid threshold, which can still emit alarms.

The scorer and structural parameters are selected using training and validation patients and are then fixed before risk calibration. Every candidate threshold is evaluated by replaying the entire causal policy, with state reset at each patient's admission. Within-patient alarm indicators are dependent, so counting them as independent calibration units is not justified. Dependence within a patient does not itself prevent calibration over independently sampled patient trajectories.

3.3. Risk Estimand and the Timestep Shortcut

For n_i decision steps, define exposure $D_i = n_i\Delta/24$ days and alarm count $A_i(\lambda) = \sum_t a_{it}(\lambda)$. The target is the patient-average daily alarm rate

$$R(\lambda) = \mathbb{E}[A_i(\lambda)/D_i]\,. \tag{2}$$

This estimand weights each patient equally and is not the pooled ward rate $\mathbb{E}[A_i]/\mathbb{E}[D_i]$, which weights patients by exposure. Both may be clinically relevant, but their identification requires different loss definitions. The present certificate concerns only Equation (2); it does not guarantee a fixed maximum count for every patient or a fixed realised total for a ward.

The synthetic comparison includes a deliberately invalid effective-sample-size shortcut. It uses the same empirical patient-average losses as the blocked selector, but substitutes the number of calibration timesteps for the number of

calibration patients in the confidence radius. This isolates the consequence of inflating the unit of information without changing the point estimator. It is not an implementation of a particular published timestep-conformal algorithm, nor does the comparison establish that every alternative time-series calibration method is invalid.

3.4. Patient-Blocked Uniform Risk Bounds

Define the patient loss as the observed daily alarm rate,

$$L_i(\lambda) = A_i(\lambda)/D_i \in [0, L_{\max}]. \tag{3}$$

The upper bound must hold over the deployment population's support, not merely over the observed calibration sample. If eligible records have at least $H_{\min}$ hours of exposure, let $n_{\min} = \lceil H_{\min}/\Delta \rceil$ and $g = \tau + 1$ without escalation, or $g = \min(\tau + 1, \kappa)$ when escalation is enabled. The minimum firing separation is g, so the exact discrete-support upper bound is

$$L_{\max} = \sup_{n \in \mathbb{N},\ n \geq n_{\min}} \frac{24}{n\Delta} \left(1 + \left\lfloor \frac{n-1}{g} \right\rfloor\right). \tag{4}$$

The rate decreases between firing-count increments, and its peaks at $n = 1+qg$ decrease as q increases; hence checking $n_{\min}$ and the next such peak is sufficient. For hourly sampling, $\tau = 4$, no escalation and $H_{\min} = 24$, a 26-hour stay yields $L_{\max} = 144/26$, exceeding the five alarms per day possible over exactly 24 hours. Escalation can increase this bound and must not be omitted from its calculation. Losses outside the declared interval are rejected, not clipped, because clipping would certify a different estimand.

Condition on independently obtained training and validation data. The scorer, structural parameters, finite threshold grid Λ, and support bound are fixed under this conditioning. Calibration subjects are independent and identically distributed draws from the same population as future deployment subjects, although each subject's trajectory may have arbitrary internal dependence. The sample size n below is the number of calibration subjects, never the number of timesteps. Exchangeability alone is not sufficient for the independent-sample concentration argument used here.

Proposition 1 (Budget validity)**.** *Let $\widehat{R}_n(\lambda) = n^{-1} \sum_{i=1}^{n} L_i(\lambda)$ and $\alpha \in (0, 1)$. Under the conditions above, select*

$$\widehat{\lambda} = \min \left\{ \lambda \in \Lambda : \widehat{R}_n(\lambda) + L_{\max} \sqrt{\frac{\log(|\Lambda|/\alpha)}{2n}} \leq B \right\} \tag{5}$$

when the set is nonempty, and select the exact no-alarm policy $\widehat{\lambda} = +\infty$ otherwise. For a finite nonnegative budget B, the selected policy satisfies

$$\Pr\{R(\widehat{\lambda}) > B\} \leq \alpha. \tag{6}$$

The probability is over calibration samples; R is a population expectation conditional on the fixed scorer and policy family.

Proof. For each fixed λ, the relevant one-sided Hoeffding inequality is

$$\Pr\{R(\lambda) - \widehat{R}_n(\lambda) > \epsilon\} \leq \exp(-2n\epsilon^2/L_{\max}^2).$$

Setting $\epsilon = L_{\max} \sqrt{\log(|\Lambda|/\alpha)/(2n)}$ and applying a union bound gives simultaneous upper bounds $R(\lambda) \leq \widehat{R}_n(\lambda) + \epsilon$ for all grid policies with probability at least $1 - \alpha$. On that event, any policy selected from the feasible set satisfies $R(\widehat{\lambda}) \leq B$. If the feasible set is empty, the fallback has identically zero loss and hence satisfies the budget without a concentration argument. Thus a budget violation can occur only on the failure event of the simultaneous upper bounds, which has probability at most α. □

The empirical-Bernstein radius used in the reported experiments has a separate finite-grid guarantee. It is not an unproved tighter variant of Proposition 1.

Proposition 2 (Empirical-Bernstein validity)**.** *Under the fixed-scorer, finite-grid and independent-subject conditions of Proposition 1, let $n \geq 2$ and define*

$$\epsilon_{\mathrm{EB}}(\lambda) = \widehat{\sigma}_n(\lambda) \sqrt{\frac{2\log(2|\Lambda|/\alpha)}{n}} + \frac{7L_{\max}\log(2|\Lambda|/\alpha)}{3(n-1)}, \tag{7}$$

where $\widehat{\sigma}_n(\lambda)$ is the sample standard deviation of $\{L_i(\lambda)\}_{i=1}^n$ with denominator $n-1$. Select

$$\widehat{\lambda}_{\mathrm{EB}} = \min\left\{\lambda \in \Lambda : \widehat{R}_n(\lambda) + \epsilon_{\mathrm{EB}}(\lambda) \leq B\right\} \tag{8}$$

when the set is nonempty, and the exact no-alarm policy otherwise. Then

$$\Pr\{R(\widehat{\lambda}_{\mathrm{EB}}) > B\} \leq \alpha. \tag{9}$$

Proof. For a fixed λ, the random variables $L_i(\lambda)$ lie in $[0, L_{\max}]$ and are independent across subjects. The empirical-Bernstein inequality of Maurer and Pontil [54] with failure probability $\delta = \alpha/|\Lambda|$ gives, with probability at least $1-\delta$,

$$R(\lambda) \leq \widehat{R}_n(\lambda) + \widehat{\sigma}_n(\lambda)\sqrt{\frac{2\log(2/\delta)}{n}} + \frac{7L_{\max}\log(2/\delta)}{3(n-1)}.$$

Substituting $\delta = \alpha/|\Lambda|$ yields the radius in (7). A union bound over the $|\Lambda|$ fixed grid points makes these inequalities simultaneous with probability at least $1-\alpha$. On that event, every policy in the feasible set of (8) satisfies $R(\lambda) \leq B$; the no-alarm fallback has zero loss. Thus a budget violation can occur only on the union-bound failure event, whose probability is at most α. □

Proposition 1 is the conservative Hoeffding certificate, while Proposition 2 covers the empirical-Bernstein certificates reported in the experiments. The latter can be tighter when the per-subject alarm counts have low variance, but it is not uniformly tighter. Both are finite-grid upper-confidence risk procedures; subject blocking does not turn either into the standard expected-risk conformal construction. The manuscript's conformal terminology denotes this broader calibration motivation, not a claim that the proof requires only conformal exchangeability.

The same fixed-grid calculation can be read as a multiple-testing problem: each candidate threshold is a null hypothesis that its risk exceeds the budget, and a valid p-value turns rejection into a certificate. This is the Learn-then-Test construction [39]. A Holm step-down correction rejects at least as many hypotheses as the Bonferroni rule in Proposition 1 while retaining family-wise error control, so it is used as a less conservative comparator with the same guarantee. Recent anytime-valid variants relax the fixed-sample requirement [40], and quantile versions target a tail rather than the mean [41]; neither is required for the patient-average estimand studied here, but both clarify that the guarantee is a property of the testing procedure rather than of the score model.

The certificate always uses the raw empirical loss at the deployed threshold. Stateful alarm counts are not assumed to be monotone in the threshold, so isotonic regression is optional descriptive output and never substitutes for that loss in the upper bound. A finite test-cohort mean can exceed the budget even when its population expectation does not, so its exceedance frequency is an empirical diagnostic rather than the exact probability in Equation (6). Likewise, calibration transferred across distributions is not covered without the same-population assumption.

3.5. Operating Points and Lead-Time Evaluation

The reference selector deploys the smallest finite threshold whose raw risk upper bound meets the requested budget. This is a deterministic operating-point convention rather than a guarantee of optimal detection. A separate utility objective could select among the uniformly certified candidates without invalidating the risk bound, but such an optimiser is not implemented here. Neither a lower threshold nor a higher budget is assumed to improve every operational metric of a stateful policy.

For an event observed during follow-up, define the detection window $\mathcal{W}_i = [\max(0, t_i^\star - W), t_i^\star)$. An event is detected if at least one alarm falls in that window, and its lead time is

$$\ell_i(\lambda) = t_i^\star - \min\{t \in \mathcal{W}_i : a_{it}(\lambda) = 1\}. \tag{10}$$

Algorithm 1 Patient-blocked upper-confidence calibration of an alarm policy

Require: independent calibration patients; fixed scorer, grid Λ, structural parameters, valid $L_{\max}$; budget B; level α

for $\lambda \in \Lambda$ **do**
    replay the causal policy on each patient's observed follow-up
    compute $L_i(\lambda)$ and reject any loss outside $[0, L_{\max}]$
    compute raw $\widehat{R}_n(\lambda)$ and Hoeffding or empirical-Bernstein radius
end for
$\mathcal{F} \leftarrow \{\lambda \in \Lambda : \widehat{R}_n(\lambda) + \epsilon(\lambda) \le B\}$
$\widehat{\lambda} \leftarrow \min \mathcal{F}$ if nonempty; otherwise $+\infty$
return threshold, raw certificate, and explicit fallback status

Lead time is undefined for undetected events and is represented as missing rather than zero. The reported median therefore describes detected events only and must always be accompanied by sensitivity and the number of eligible events. Events at admission, outside observed follow-up, or without an identifiable onset cannot supply an observed pre-onset lead time. A retrospective top-k selection over a complete trajectory is noncausal and may serve only as a labelled oracle comparator, not as a deployable early-warning policy.

Budget sweeps compare methods under the same requested upper-risk constraint, not necessarily at identical realised alarm rates. The plotted frontier uses realised alarm rate and conditional median lead time, retaining only nondominated pairs. At equal alarm rates only the largest lead time is retained; at larger rates a point remains only when its lead time strictly improves. Because conditional lead time can improve by missing difficult cases, sensitivity must be reported alongside the frontier before any clinical interpretation.

4. Experimental Setup

4.1. Data Sources and Cohort Audit

Figure 1 summarizes the evaluation design. We use the public PhysioNet/CinC Challenge 2019 cohort [55], which contains 20,336 records from site A and 20,000 records from site B. The two sites are treated as distinct hospitals rather than interchangeable shards. All records were downloaded from the public release and checked against the published object sizes and MD5 entities before parsing. Site A contains 1,790 records with a positive sepsis label and 203 left-censored records whose first observed label is already positive; site B contains 1,142 positive and 223 left-censored records. After the six-hour label shift described below, 1,587 A records and 919 B records have an identifiable onset inside observed follow-up. Median observed stay is 39 hours at A and 38 hours at B.

The audit also exposes substantial measurement heterogeneity. In site A, $EtCO_2$ is absent in every record, whereas 7.6% of B records contain at least one $EtCO_2$ value; temperature is observed in only 33.8% of A records, and several laboratory variables are observed in fewer than 5%. This pattern motivates retaining a fixed variable set across sites and explicitly handling all-missing training columns, rather than fitting site-specific feature matrices. The 2012 mortality cohort [56] was retained only as an ingestion and outcome-join check; because it has no deterioration onset, we do not treat its 49 decision points as a same-task external validation.

4.2. Labels, Partitions and Eligibility

The supplied 2019 label becomes positive six hours before the challenge-defined clinical onset. We set `label_lead_hours=` 6 to recover the candidate clinical time and `horizon=` 6 to label the six preceding hours positive. Timesteps at or after onset, padded steps, and all steps of a left-censored record are excluded from discrimination and calibration targets. An event at or beyond the end of observed follow-up is not counted as a detectable lead-time case.

The official 2019 release stores one PSV file per subject, and we use "subject" and "patient" interchangeably for this cohort. We use the file stem as the subject identifier and verify that every identifier occurs in exactly one partition; the split audit is written to the run metadata. For each random seed we use a subject-disjoint 60/15/10/15 split into training, validation, calibration and test subjects. The scorer, preprocessing, structural policy parameters and threshold grid are fixed using training and validation subjects. The calibration partition selects the deployed threshold,

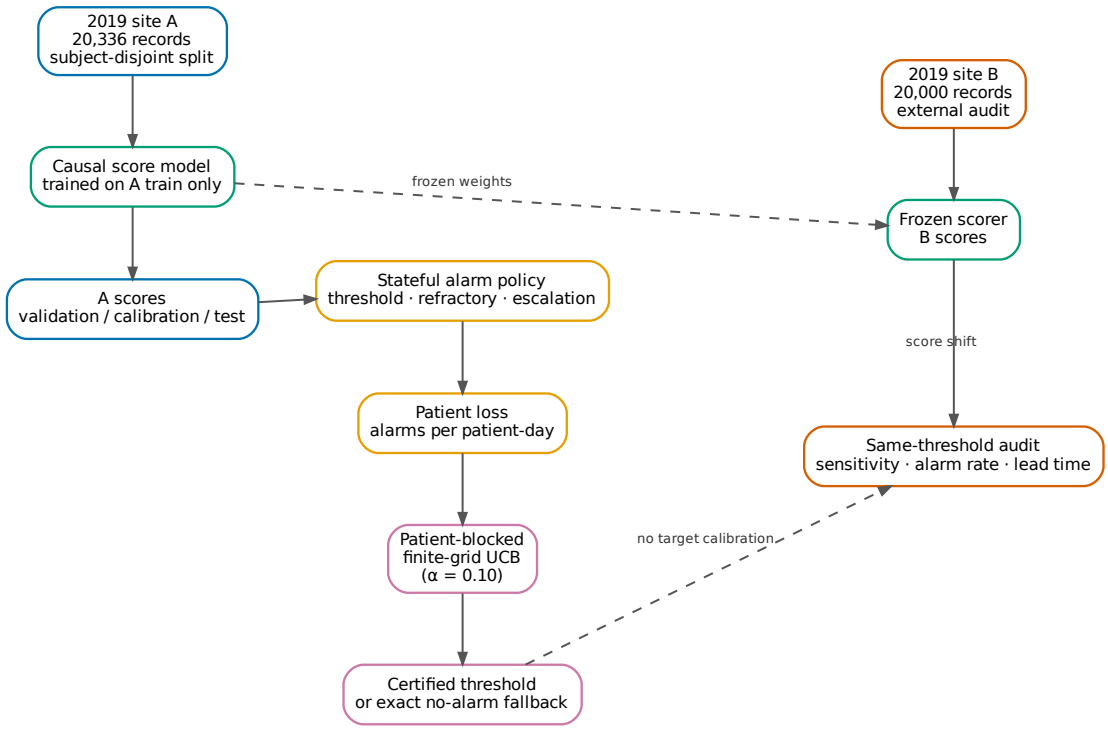


Figure 1: Evaluation design. A causal score model is fitted on site A training records and frozen before site B is touched. The stateful policy replays each patient trajectory to obtain alarms per patient-day; a patient-blocked finite-grid upper confidence bound selects a threshold, or returns the exact no-alarm fallback when no candidate is certified. The primary external audit applies the same threshold without target-site calibration; the target-calibration diagnostic in the Results is a separate oracle analysis.

and the test partition is touched only for evaluation. Site B is evaluated with the A-trained scorer and the A-calibrated threshold; no B labels are used for target calibration. Three seeds (0, 1, 2) are reported for the prefix logistic model, the masked GRU, the LightGBM baseline and the E2 budget sweep. The target-site recalibration diagnostic is reported for all three logistic seeds and for one GRU seed, and is labelled as an oracle analysis throughout.

A separate target-site diagnostic asks how much of the external loss is recoverable when target labels are available. We split the eligible B subjects into disjoint calibration and test halves of equal size using a site-specific random permutation (seed $1000 + s$ for source seed s), recalibrate the same policy family and threshold grid on B calibration subjects, and evaluate on B test subjects. This analysis uses target labels for threshold selection and is therefore an oracle-style target-calibrated reference estimate rather than a deployable label-free procedure; it is reported separately from the primary external audit and never used to tune the A model.

The target label-budget curve fixes the B test half and draws nested calibration subsets of 5%, 10%, 25% and 50% of all eligible B subjects from the remaining half. The same threshold selector is refitted on each subset and evaluated on the common test half, so the curve isolates calibration-set size. The source-threshold point is evaluated on the same test half and serves as the zero-label reference.

The support bound requires an independently declared minimum exposure. We use $H_{\min} = 24$ hours, which retains 15,837 of 20,336 A records and 14,824 of 20,000 B records. This is an eligibility restriction, not a post-hoc filter chosen to improve results. Alarm-rate evaluation counts the full observed exposure, including time after onset; lead-time evaluation uses only the pre-onset window. With hourly decisions, refractory $\tau = 4$ and no escalation, the resulting support bound is $L_{\max} = 144/26 \approx 5.54$ alarms per patient-day.

4.3. Score Models

The primary scorer is a causal prefix-feature logistic regression. At each decision step it uses the last observed value of every variable, the cumulative observation fraction, and admission static features. All features are computed from current and past observations only. Median imputation is fitted on the training partition with empty features retained, so a variable that is entirely missing at site A remains a constant column at site B instead of silently changing the feature dimension. The classifier uses $C = 1$ and a maximum of 2,000 iterations. A unidirectional masked GRU with 64 hidden units is used as a secondary architecture check. It consumes observed values, observation masks, log-transformed elapsed time and static features; it is a reference implementation, not a GRU-D reproduction. The GRU is trained for 20 epochs with Adam at 10^{-3}, batch size 64, and a fixed random seed on an RTX 4090. A LightGBM gradient-boosted tree classifier is fitted to the same prefix features with 300 trees, learning rate 0.05, 31 leaves and median imputation; it is reported as a stronger tabular baseline, not as a tuned state-of-the-art model.

Partial NEWS, SIRS and qSOFA scores are computed from the variables available in the 2019 release, following the standard early-warning-score and instrument definitions [30, 57]. NEWS omits the consciousness and supplemental-oxygen components, SIRS omits the immature-neutrophil alternative, and qSOFA omits the Glasgow Coma Scale; these are research-only partial instruments, not certified clinical implementations. Charted observations are placed on the ICULOS clock, forward-filled, and scored per timestep. Steps before the first observed component receive zero points rather than being backfilled from a later observation, so the policy cannot use future measurements; the same no-points convention is applied to other unobserved steps. This is a missing-data convention, not an assertion that the patient was clinically normal. Their thresholds are calibrated on the same A calibration subjects at the same requested budgets as the learned scorers, so the comparison is a matched-workload audit rather than a claim that the instruments are equivalent.

4.4. Alarm Policy and Calibration

The policy family is the stateful rule in Equation (1). We fix refractory $\tau = 4$ decision steps, escalation $\kappa = 0$, and a 12-hour lead window $W = 12$. The candidate grid is 200 equally spaced thresholds from 0.01 to 0.99; the grid is fixed before calibration. The risk level is $\alpha = 0.10$, and the reported certificates use the empirical-Bernstein radius in Equation (7); Proposition 2 gives its finite-grid validity, while the Hoeffding radius in Proposition 1 is retained as the conservative variant. The selector deploys the smallest threshold whose raw empirical loss plus radius is at most the requested budget. If no threshold is certified, the exact no-alarm fallback is reported rather than silently substituting the largest finite threshold.

For each patient and threshold, the full stateful trajectory is replayed once to obtain alarms per patient-day. This loss matrix is reused across budgets and across the blocked and timestep procedures, so differences between procedures come only from the confidence radius and not from a different point estimator. The deliberately invalid timestep comparator substitutes calibration timesteps for calibration patients in the Hoeffding radius; it is a diagnostic, not a proposed method.

The finite-grid certificate is compared with four alternatives on the same loss matrix. A plug-in rule selects the smallest threshold whose empirical loss is at most the budget and has no finite-sample correction. A Learn-then-Test selector converts each candidate into a one-sided test and applies Holm or Bonferroni correction [39]. A randomised mixture draws one of two adjacent thresholds per patient; the mixture is certified over the augmented candidate family and can use the budget more fully than a single grid point. A validation-selected variant restricts attention to certified thresholds and ranks them by sensitivity on a disjoint validation partition. The certificate is always computed on calibration patients; validation is used only to break ties among already-certified candidates.

Repeated-split coverage is estimated with ten stratified partitions of the eligible cohort on each site, holding event and non-event patients separate. For each budget and procedure we report the fraction of splits whose realised mean rate exceeds the budget, the fallback rate, budget utilisation, sensitivity and the operational metrics. These split indicators estimate finite-cohort behaviour; they do not prove the population theorem, and their Monte Carlo resolution is reported alongside the point estimate.

Label-free target adaptation matches the requested rate on a disjoint warm-up half of the target site: the most aggressive threshold whose empirical rate is at most the budget is deployed, and sensitivity is evaluated on the held-out target half. This uses no target labels for threshold selection and is therefore a deployable heuristic, not a coverage guarantee. A ward simulation resamples observed per-hour alarm traces with Poisson arrivals, finite beds and a

Table 1: Public 2019 cohort audit after downloading and checksum verification. Percentages for observed variables are the fraction of records with at least one non-missing value. The two sites are separate hospital systems.

Quantity	Site A	Site B
Records	20,336	20,000
Positive records	1,790	1,142
Left-censored records	203	223
Identifiable pre-onset events	1,587	919
Median observed stay (h)	39	38
$EtCO_2$ observed (%)	0.0	7.6
Temperature observed (%)	33.8	33.9
Lactate observed (%)	3.4	1.9

fixed hourly alarm-handling capacity; it translates per-patient rates into peak demand and the fraction of hours above capacity. One-at-a-time ablations vary the refractory window, escalation tier, lead window, minimum stay, risk level and threshold-grid resolution around the primary configuration.

4.5. Metrics and Uncertainty

Discrimination is summarized by AUROC and AUPRC over usable pre-onset timesteps. Because the positive-timestep prevalence is only 1.27% in the A test partition and 0.72% at B, AUPRC is interpreted against that prevalence rather than against 0.5. Calibration is summarized by 10-bin expected calibration error (ECE); ECE is descriptive and does not establish clinical calibration. Operational metrics are the realised patient-average alarm rate, event sensitivity in the 12-hour lead window, median lead time among detected events, the fraction of test patients whose per-day alarm count exceeds the requested budget, positive predictive value (PPV), and the number needed to alert (NNA).

Confidence intervals for score metrics use a patient-level bootstrap with 400 resamples, preserving within-patient dependence. Operational intervals use 2,000 patient-block resamples on seed 0 and resample whole target subjects, so within-subject alarm dependence is preserved; paired differences use the same resampled subjects and Holm correction across the reported comparisons. Two-sided bootstrap p-values are not a substitute for external validation. Operational curves report the mean and sample standard deviation over seeds; no interval is attached to the single-seed target-site GRU diagnostic.

4.6. Reproducibility and Compute

All raw records and score caches are kept outside the manuscript repository. The code exposes separate entry points for downloading and checksumming the 2019 release, auditing the official files, fitting and caching E1 scores, sweeping E2 budgets, bootstrap intervals and seed aggregation. The full test suite (164 tests) passes on the execution machine. The GRU uses CUDA but no formal GPU-hour or energy measurement was recorded; runtimes are therefore reported as implementation facts, not as a compute-efficiency claim.

5. Results

5.1. Cohort Audit and Score Discrimination

The public cohort is large enough to estimate patient-level alarm rates, but its measurement process is far from uniform. Table 1 shows that site A has a higher positive-record fraction than site B (8.8% versus 5.7%), while both sites contain a non-trivial number of left-censored records whose first label is already positive. The observed-variable table also explains why a naive cross-site model can fail: $EtCO_2$ is entirely absent at A but present in 7.6% of B records, and temperature is observed in only about one third of either cohort. A model that drops an all-missing training column would therefore change its input dimension at the external site; the implementation retains such a column as a constant during imputation and the regression test in the repository checks this case explicitly.

Table 2 reports three-seed score discrimination. The prefix logistic model reaches an A-test AUROC of 0.736 ± 0.013 and AUPRC of 0.034 ± 0.006; the masked GRU reaches 0.743 ± 0.011 and 0.043 ± 0.005, and LightGBM reaches

Table 2: Three-seed score discrimination (mean ± sample standard deviation). AUPRC is computed over usable pre-onset timesteps; positive-timestep prevalence is 1.27% at A and 0.72% at B. Ten-bin ECE is reported in the text for all three models and is descriptive.

	Site A test		Site B external	
Model	AUROC	AUPRC	AUROC	AUPRC
Prefix logistic	0.736 ± 0.013	0.034 ± 0.006	0.676 ± 0.005	0.0172 ± 0.0002
Masked GRU	0.743 ± 0.011	0.043 ± 0.005	0.683 ± 0.004	0.0183 ± 0.0016
LightGBM	0.763 ± 0.009	0.042 ± 0.004	0.713 ± 0.003	0.0250 ± 0.0010

0.763 ± 0.009 and 0.042 ± 0.004. These AUPRC values must be read against the 1.27% positive-timestep prevalence: both models rank positive steps several times better than chance, but the absolute positive predictive signal remains modest. On site B the logistic model falls to 0.676 ± 0.005 AUROC and 0.017 ± 0.000 AUPRC, while the GRU reaches 0.683 ± 0.004 and 0.018 ± 0.002, and LightGBM reaches 0.713 ± 0.003 and 0.025 ± 0.001. Ten-bin ECE is small in absolute terms (0.0011, 0.0017 and 0.0064 at A; 0.0014, 0.0018 and 0.0062 at B for logistic, GRU and LightGBM, respectively), but it is descriptive and should not be read as evidence of clinical calibration under shift. The drop is consistent with multisite sepsis-model evaluations that report degraded transport across hospitals [36, 37], and it is larger in AUPRC than in AUROC because prevalence also changes.

The seed-0 patient bootstrap sharpens the comparison without turning a small numerical difference into a claim. At site A the logistic model has AUROC 0.741 with 95% interval [0.710, 0.766] and AUPRC 0.041 with interval [0.035, 0.051]; the GRU has AUROC 0.744 [0.716, 0.772] and AUPRC 0.044 [0.038, 0.056]. The paired GRU-minus-logistic difference is 0.004 for AUROC with interval [−0.023, 0.032] and two-sided $p = 0.775$, and 0.003 for AUPRC with interval [−0.006, 0.014] and $p = 0.42$. The GRU is therefore a secondary architecture check, not a demonstrated improvement. At site B the AUROC difference is essentially zero, while the AUPRC difference is 0.003 with interval [0.0002, 0.0059] and $p = 0.05$; this isolated interval is reported as exploratory because it is not corrected across metrics or sites.

5.2. Budgeted Alarm Policies and Cross-Site Transport

Figure 2 and Table 3 report the operational frontier for the prefix logistic scorer. On site A, raising the requested budget from 0.25 to 2 alarms per patient-day increases mean sensitivity from 0.185 ± 0.061 to 0.774 ± 0.032, while the realised alarm rate rises from 0.108 ± 0.012 to 1.366 ± 0.026. The realised rate stays below the requested budget at every operating point, which is the expected consequence of a finite-sample upper bound rather than evidence that the policy is slack by design. The tightest budget has the largest seed-to-seed sensitivity variance, because a small change in the calibrated threshold moves several of the 168 eligible A events across the detection boundary. Median lead time among detected events stays between 8.7 and 10 hours, but this conditional median must be read with the corresponding sensitivity: it describes only the events that were detected.

The external audit exposes a qualitatively different operating point. At the 1-per-day budget the A-calibrated policy emits 0.276 ± 0.063 alarms per B patient-day, less than half the A rate, and detects 0.315 ± 0.062 of eligible B events compared with 0.585 ± 0.020 at A. At the tightest budget, B sensitivity falls to 0.081 ± 0.026 while the realised B alarm rate is only 0.040 ± 0.014. The direction of the rate change is not a free lunch: the same threshold is simply less active on the shifted score distribution, and the lost activity is concentrated on true events. This is exactly the failure mode that a threshold-only transport claim would hide, and it is consistent with recent evidence that score distributions and calibration can shift across hospitals [45, 58]. A target-site adaptation step is therefore necessary before deployment, and the label-free rate-matching study below evaluates one such step without target labels.

The no-recalibration audit establishes that the source threshold does not travel, but it does not quantify how much of the loss is recoverable by adaptation. To estimate that quantity, we split target B into subject-disjoint calibration and test halves (7,412 patients each; on average 354 calibration events and 337 test events over three seeds), recalibrated the same policy family on B calibration labels, and evaluated it on the B test half. This is an oracle diagnostic: target labels are used for threshold selection, so it is not a deployable label-free procedure and must not be compared with the primary external audit as if it were one. Table 4 reports the result for the logistic scorer. At 1 alarm per patient-day, target calibration raises mean B sensitivity from 0.294 ± 0.082 to 0.497 ± 0.022 and the realised rate from 0.270 ± 0.065 to 0.616 ± 0.058; at 0.25 per day it raises sensitivity from 0.073 ± 0.026 to 0.218 ± 0.036 at a realised

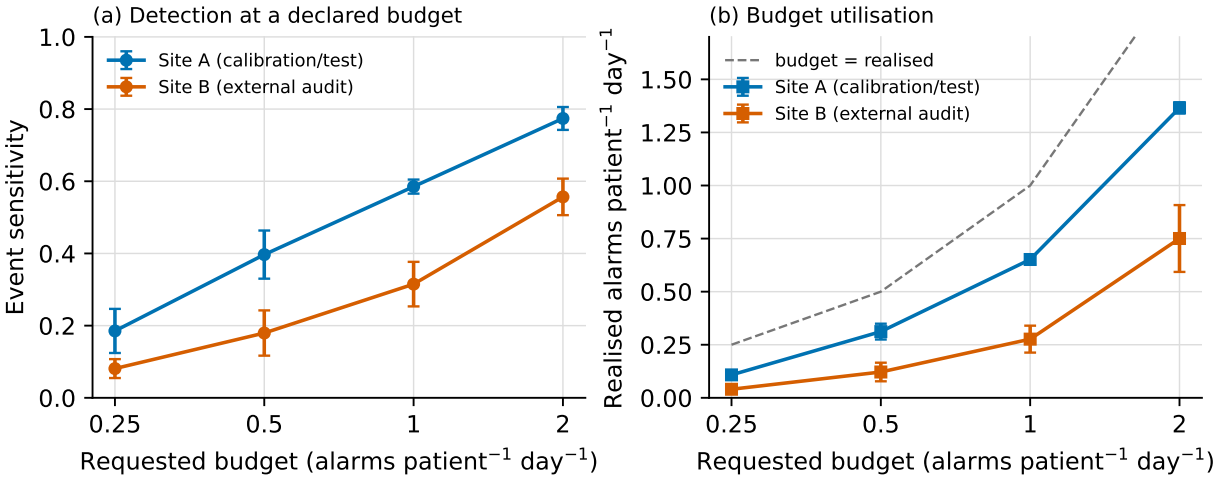


Figure 2: Three-seed E2 operating curves for the prefix logistic scorer. (a) Event sensitivity versus the requested patient-average alarm budget. (b) Realised alarm rate versus the requested budget; the dashed diagonal marks exact budget utilisation. Error bars are sample standard deviations over seeds 0–2. Site B uses the A-calibrated threshold without target labels.

rate of 0.159 ± 0.009. The recovery is real but partial, and it is purchased with a higher alarm load and a higher patient-violation fraction because the target-calibrated threshold is lower. The single-seed GRU diagnostic shows the same direction (0.314 to 0.493 at 1 per day). At 2 per day the target-recalibrated selector hits the threshold-grid floor: sensitivity saturates at 0.724 ± 0.056 while the realised rate rises to 1.466 ± 0.260, so the diagnostic identifies a range where adaptation matters rather than a universal gain.

The timestep comparator is a diagnostic, not a proposed method, and the repeated-split study in Section 5.7 changes how it should be read. In the single split it selects a lower threshold than the blocked certificate at some budgets and raises sensitivity, but its realised mean rate remained below the requested budget in all ten repeated splits on both sites. It behaves like the plug-in rule because its effective sample size is large. The defensible criticism is therefore not that the shortcut always violates the mean-rate constraint; it is that the shortcut supplies no patient-level guarantee, and at 1 per day its patient-violation fraction is 0.296 versus 0.210 for the blocked certificate. A mean-rate bound estimated from correlated hours cannot certify the patient-average estimand, even when the realised mean happens to stay below budget on one cohort.

The GRU changes the level of the frontier but not its transport behaviour. At 1 alarm per patient-day its three-seed mean A sensitivity is 0.684 ± 0.006 and its B sensitivity is 0.375 ± 0.078, compared with 0.585 ± 0.020 and 0.315 ± 0.062 for the logistic model; the B alarm rate is also higher (0.328 ± 0.164 versus 0.276 ± 0.063). At 0.25 per day the GRU reaches 0.301 ± 0.082 at A and 0.115 ± 0.027 at B, again above the logistic frontier. The GRU is therefore a secondary architecture check rather than a demonstrated improvement: its seed-level spread is large, its paired site-B sensitivity interval includes zero after Holm adjustment, and its external alarm rate is higher. LightGBM is stronger on site B but, as Section 5.4 shows, it still leaves the realised rate below the requested budget and does not remove the need for target calibration. A more expressive scorer can move the external frontier without repairing the site shift, and it can raise external alarm load while doing so. The calibration procedure, not the backbone alone, determines whether a stated alert budget is met.

Two limitations of the grid are visible in the table. First, the 4-per-day row already sits at the threshold grid floor and, in the full sweep, coincides with 8 per day; the grid stops at 0.01; above roughly 2 alarms per patient-day the budget constraint is no longer binding and the frontier is truncated by the candidate set. Second, every certified operating point has a realised rate below its requested budget, so the experiment demonstrates conservative risk control rather than tight budget utilisation. Neither observation undermines the certificate, but both matter for engineering deployment: a hospital that needs a precise alarm load may prefer a finer grid or a utility-based selection among certified thresholds, and a hospital that needs a higher sensitivity must accept the corresponding increase in realised alert volume. The repeated-split study in Section 5.7 shows that the empirical-Bernstein LTT and randomised-mixture

Table 3: E2 budget sweep for the prefix logistic scorer, mean ± sample standard deviation over three seeds. Rates are alarms per patient-day; lead time is the median among detected events in hours; violation is the fraction of test patients whose per-day count exceeds the budget. The 4-per-day row is at the threshold grid floor; in the full sweep it coincides with 8 per day.

Site A (calibration/test)				
Budget	Rate	Sens.	Lead	Viol.
0.25	0.108 ± 0.012	0.185 ± 0.061	9.5 ± 0.5	0.074 ± 0.001
0.50	0.312 ± 0.037	0.397 ± 0.067	8.7 ± 0.6	0.151 ± 0.014
1.00	0.651 ± 0.022	0.585 ± 0.020	9.3 ± 0.6	0.210 ± 0.003
2.00	1.366 ± 0.026	0.774 ± 0.032	10.0 ± 0.0	0.309 ± 0.004
4.00	2.096 ± 0.026	0.894 ± 0.011	10.0 ± 0.0	0.271 ± 0.012

Site B (external audit)				
Budget	Rate	Sens.	Lead	Viol.
0.25	0.040 ± 0.014	0.081 ± 0.026	8.5 ± 0.5	0.030 ± 0.010
0.50	0.122 ± 0.044	0.180 ± 0.063	8.2 ± 0.3	0.065 ± 0.023
1.00	0.276 ± 0.063	0.315 ± 0.062	8.3 ± 0.6	0.094 ± 0.025
2.00	0.750 ± 0.157	0.557 ± 0.051	9.0 ± 0.0	0.163 ± 0.038
4.00	1.473 ± 0.250	0.724 ± 0.049	9.3 ± 0.6	0.131 ± 0.025

Table 4: Target-site recalibration diagnostic for the prefix logistic scorer, mean ± sample standard deviation over three source seeds. The target hospital is split 50/50 into subject-disjoint calibration and test halves; the calibration half uses target labels to select the threshold. Sensitivity and realised alarm rate are evaluated on the target test half (7,412 patients, about 337 events). This is an oracle-style target-calibrated reference estimate, not a deployable label-free procedure.

	No recalibration		Target recalibrated	
Budget	Sens.	Rate	Sens.	Rate
0.25	0.073 ± 0.026	0.039 ± 0.014	0.218 ± 0.036	0.159 ± 0.009
0.50	0.173 ± 0.074	0.118 ± 0.043	0.358 ± 0.031	0.360 ± 0.019
1.00	0.294 ± 0.082	0.270 ± 0.065	0.497 ± 0.022	0.616 ± 0.058
2.00	0.544 ± 0.072	0.744 ± 0.165	0.724 ± 0.056	1.466 ± 0.260

selectors can raise utilisation without a mean-rate violation, so the conservatism of the blocked row is a property of that selector rather than of the budget constraint itself. The clinical interpretation of these rates also depends on local workflow and alarm-fatigue tolerance, which this retrospective study does not measure [7, 8].

5.3. How Many Target Labels Are Needed?

The single 50/50 target diagnostic answers whether adaptation can recover sensitivity, but not how much target annotation that recovery requires. We therefore fix the target test half and draw nested calibration subsets from the remaining target pool at 5%, 10%, 25% and 50% of all eligible target subjects. Each subset is drawn without replacement from the same shuffled pool, so the curve isolates calibration-set size rather than a change of test patients. The source threshold is evaluated on the same test half as the no-recalibration reference, and every recalibrated threshold is selected by the same patient-blocked procedure used elsewhere.

Table 5 reports the resulting label-budget curve for the logistic scorer, averaged over the three source seeds. At the 1-per-day budget, 741 target patients (5% of the eligible cohort, about 34 calibration events) raise mean test sensitivity from 0.294 ± 0.082 to 0.443 ± 0.106, which is most of the 0.497 ± 0.022 obtained with the full calibration pool. At 10% (1,482 patients, about 72 events), the curve reaches the full-pool value, and additional labels do not change the selected threshold on these seeds. The lower budget behaves differently: at 0.5 per day, 5% reaches 0.254 ± 0.066 against a full-pool value of 0.358 ± 0.031, 25% reaches 0.329 ± 0.046, and only the largest subset closes the gap. A hospital that needs a low alarm load therefore needs more target labels than one operating at 1 per patient-day, because the threshold sits in a denser part of the score distribution where small calibration errors move more patients across the boundary.

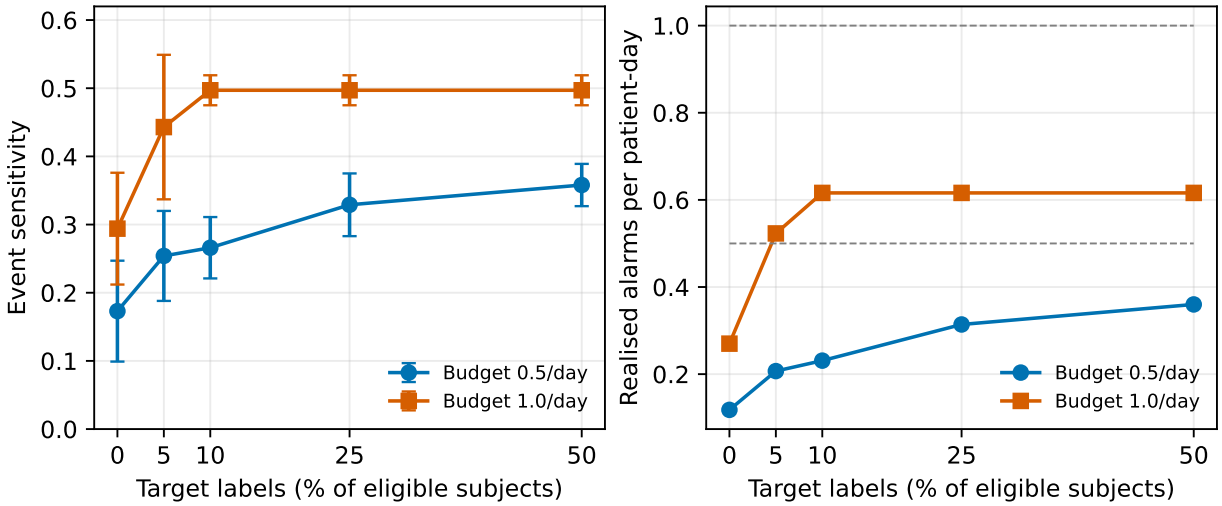


Figure 3: Target label-budget curve for the logistic scorer, averaged over three source seeds. Left: event sensitivity versus the percentage of eligible target subjects used for threshold calibration; error bars are sample standard deviations over seeds. Right: realised alarms per patient-day against the requested budget (dashed lines), showing that label cost and operating point interact. The zero-label point applies the source threshold without target calibration.

Table 5: Target label-budget curve for the prefix logistic scorer, mean ± sample standard deviation over three source seeds. The target test half contains 7,412 patients and about 337 events. "None" applies the source threshold without target labels. The remaining rows recalibrate on a nested subset of the target calibration pool; labels and rates are evaluated on the same test half. This is an oracle-style label-budget diagnostic, not a deployable label-free procedure.

			Budget 0.5		Budget 1.0	
Labels	Patients	Events	Sens.	Rate	Sens.	Rate
None	0	0	0.173 ± 0.074	0.118	0.294 ± 0.082	0.270
5%	741	34	0.254 ± 0.066	0.207	0.443 ± 0.106	0.523
10%	1,482	72	0.266 ± 0.045	0.231	0.497 ± 0.022	0.616
25%	3,706	176	0.329 ± 0.046	0.314	0.497 ± 0.022	0.616
50%	7,412	354	0.358 ± 0.031	0.360	0.497 ± 0.022	0.616

Two cautions follow from the table. First, the 5% point has the widest seed-to-seed spread at 1 per day (±0.106 versus ±0.022 for the full pool), so a small target set can recover the mean operating point while remaining unstable across source models. Second, the curve is conditional on the target split and on the policy family; it measures threshold-selection label cost, not the cost of retraining the scorer. The result is an engineering signal about the label cost of a *certified* target threshold; the label-free rate-matching rule in Section 5.9 can recover the operating point on this cohort without target labels, but it controls the alarm rate rather than the risk and therefore does not replace this curve when a finite-sample guarantee is required.

5.4. Backbone Comparison at a Fixed Alarm Budget

A budget curve is only as useful as the scores it ranks. Table 6 therefore compares the three backbones at the 1-per-patient-day operating point, holding the policy family, support bound and threshold grid fixed. At site A the GRU has the highest mean sensitivity (0.684 ± 0.006), followed by LightGBM (0.661 ± 0.042) and the prefix logistic model (0.585 ± 0.020). At site B the ordering changes: LightGBM reaches 0.540 ± 0.071, the GRU 0.375 ± 0.078, and the logistic model 0.315 ± 0.062. The LightGBM advantage is not a free lunch, because its B alarm rate is also higher (0.410 ± 0.156 versus 0.276 ± 0.063 for logistic and 0.328 ± 0.164 for the GRU), but all three policies remain below the requested B budget of 1 per day. A stronger tabular scorer therefore narrows the transport gap without eliminating it.

Table 6: Backbone comparison at the 1-per-patient-day operating point, mean ± sample standard deviation over three seeds. The threshold is calibrated on site A and applied unchanged to site B. Rates are realised alarms per patient-day; the requested budget is 1.0. LightGBM is a stronger tabular baseline, not a tuned state-of-the-art model.

	Site A		Site B	
Model	Sens.	Rate	Sens.	Rate
Prefix logistic	0.585 ± 0.020	0.651 ± 0.022	0.315 ± 0.062	0.276 ± 0.063
Masked GRU	0.684 ± 0.006	0.635 ± 0.121	0.375 ± 0.078	0.328 ± 0.164
LightGBM	0.661 ± 0.042	0.655 ± 0.147	0.540 ± 0.071	0.410 ± 0.156

Table 7: Patient-block bootstrap at the 1-per-patient-day operating point on site B, seed 0, 2,000 resamples. Intervals are 95% percentile intervals. ΔSens is the paired difference against the blocked prefix-logistic policy; p_{Holm} is adjusted across the reported rows and metrics. The global rows use a point-estimate threshold and are workload comparators, not certified policies.

Policy	Sens. [95% CI]	Rate [95% CI]	ΔSens [95% CI]	p_{Holm}
Prefix logistic, blocked	0.382 [0.348, 0.417]	0.348 [0.333, 0.362]	–	–
Prefix logistic, global	0.482 [0.443, 0.519]	0.549 [0.531, 0.568]	+0.100 [0.049, 0.152]	0.040
Masked GRU, blocked	0.326 [0.291, 0.359]	0.209 [0.198, 0.219]	−0.056 [-0.106, -0.009]	0.288
Masked GRU, global	0.408 [0.370, 0.445]	0.343 [0.329, 0.357]	+0.026 [-0.029, 0.076]	1.000
LightGBM, blocked	0.463 [0.427, 0.499]	0.260 [0.250, 0.272]	+0.081 [0.027, 0.133]	0.066
LightGBM, global	0.537 [0.501, 0.574]	0.400 [0.387, 0.413]	+0.155 [0.103, 0.208]	0.033

The seed-0 patient-block bootstrap makes the LightGBM–logistic contrast concrete without turning it into a universal ranking. At site B and 1 per day, the paired LightGBM-minus-logistic sensitivity difference is 0.081 with 95% interval $[0.027, 0.133]$ and Holm-adjusted $p = 0.066$; the alarm-rate difference is -0.087 with interval $[-0.105, -0.069]$ and adjusted $p = 0.036$, because LightGBM selects a lower threshold and alarms less often on the shifted distribution. On site A the same contrast is not significant after correction, so the external comparison is the relevant one. The sensitivity advantage is consistent but not decisive after correction, while the lower alarm rate is clearer. The result supports a modest claim: backbone quality matters for transport, but it changes the level of the frontier rather than removing the need for target calibration. A hospital still cannot infer its realised alarm load from an A-calibrated threshold alone.

5.5. Patient-Block Uncertainty

Seed-to-seed standard deviations describe calibration variability, not sampling uncertainty on the target population. We therefore resample whole B subjects with replacement 2,000 times at the 1-per-day operating point and recompute sensitivity, realised rate and patient-violation fraction on each replicate. Resampling subjects rather than hours preserves the within-patient dependence that the calibration argument is built on. Table 7 reports the seed-0 intervals and paired differences against the blocked prefix-logistic policy; Holm adjustment covers the six model–procedure rows and three metrics.

Three patterns survive the intervals. First, the blocked prefix policy is conservative on this target: its mean rate is 0.348 with interval $[0.333, 0.361]$, well below the requested 1 per day, and its sensitivity interval is $[0.348, 0.417]$. Second, the global point-estimate threshold raises sensitivity by 0.100 but also raises the realised rate by 0.201 alarms per patient-day and the patient-violation fraction from 0.122 to 0.190; the gain is a change of operating point, not a free improvement in ranking. Third, LightGBM's blocked policy reaches 0.463 sensitivity at 0.260 alarms per day, a paired gain of 0.081 over the blocked logistic policy with interval $[0.027, 0.133]$; the Holm-adjusted $p = 0.066$ is not significant at the 0.05 level, so this is a consistent but not decisive advantage. The GRU difference is negative at this operating point and is also not significant after Holm adjustment, which is why the GRU is presented as a secondary check rather than a demonstrated improvement.

These intervals are conditional on one target split and one seed; they quantify target sampling noise, not source-seed or hospital-level variability. They also do not cover the oracle target-calibration analysis, which uses target labels and is reported separately. The practical reading is therefore limited but useful: at a fixed 1-per-day request, the choice of backbone and threshold procedure can move B sensitivity by roughly ten percentage points, while the realised rate remains below the request for every certified row in the table.

Table 8: Partial clinical-instrument baselines at the 1-per-patient-day operating point. Thresholds are calibrated on site A and applied unchanged to site B. NEWS, SIRS and qSOFA omit components unavailable in the 2019 release and are research-only partial instruments. qSOFA has a three-level grid, so its realised rate is far from the request.

	Site A		Site B	
Policy	Sens.	Rate	Sens.	Rate
Prefix logistic	0.585	0.651	0.315	0.276
LightGBM	0.661	0.655	0.540	0.410
Partial NEWS	0.399	0.730	0.449	0.586
Partial SIRS	0.470	0.796	0.424	0.493
Partial qSOFA	0.220	0.492	0.240	0.327

5.6. Partial Clinical-Instrument Baselines

A learned score is not the only way to operate an alarm policy, so we calibrated partial NEWS, SIRS and qSOFA on the same A calibration subjects and evaluated them at the same requested budgets. Table 8 reports the 1-per-day operating point. The instruments are coarse integer scores, and the calibration grid is limited to their attainable values; qSOFA, for example, has only three levels, so the selector can miss a requested budget by a wide margin or return the no-alarm fallback. On site A, NEWS reaches 0.399 sensitivity at 0.730 alarms per day, SIRS reaches 0.470 at 0.796, and qSOFA reaches only 0.220 at 0.492. The learned logistic policy is more sensitive at a lower rate (0.585 at 0.651), and LightGBM is higher still (0.661 at 0.655).

The external ordering is not the same as the internal one. NEWS and SIRS beat the source-threshold logistic policy on site B (0.449 [0.412, 0.484] and 0.424 [0.389, 0.461] versus 0.315), but they do so at higher realised rates (0.586 [0.570, 0.602] and 0.493 [0.478, 0.508] versus 0.276), and neither reaches the LightGBM sensitivity of 0.540 at 0.410. The paired sensitivity differences against the blocked logistic policy are +0.067 [0.016, 0.118] for NEWS, +0.042 [-0.008, 0.093] for SIRS and −0.142 [-0.190, -0.095] for qSOFA; only the qSOFA loss survives Holm correction, and all three clinical instruments increase the realised rate. This is a workload comparison, not evidence that the instruments are clinically equivalent: their missing components are systematic, their scores are not probabilities, and their thresholds were selected on A. The result does show that a simple, auditable instrument can be a competitive external baseline once the comparison is made at a declared patient-level alarm budget instead of at an arbitrary threshold.

The coarse grids also expose a practical failure mode that probability-valued scorers hide. At 0.25 per day, qSOFA has no admissible threshold and returns the no-alarm fallback on both sites; at 0.5 per day its only feasible threshold alarms on 0.492 of A patient-days and 0.327 of B patient-days. NEWS and SIRS similarly jump between adjacent integer levels, which makes the realised rate a step function of the request. A hospital that needs fine workload control should therefore not rely on an integer instrument alone, even when its discrimination is adequate; the policy layer needs either a richer score or an explicit randomization/utility rule over neighbouring thresholds. These baselines are reported to bound what a simple instrument can do under the same budget protocol, not to propose one for deployment.

5.7. Repeated-Split Coverage and Budget Utilisation

A single calibration/test split cannot estimate how often the certificate is violated. We therefore repeat the patient-level split ten times on each site, calibrate on half of the eligible subjects and evaluate on the other half, and record the realised mean rate, the fraction of splits above the requested budget, the no-alarm fallback rate and the patient-level violation fraction. Figure 4(a) plots budget utilisation; Table 9 reports the 1-per-day row.

Three findings follow. First, no certified selector exceeded the mean budget in any of the ten splits on either site; the plug-in rule exceeded it in one of ten splits at the 0.25-per-day budget, where its empirical rate was 0.221 against a request of 0.25 on average but crossed the line in a single partition. Second, the timestep shortcut also stayed below the mean budget in all splits. The data therefore do not support the stronger claim that the shortcut always violates the mean-rate constraint; its deficiency is that it has no patient-level guarantee, and at 1 per day its patient-violation fraction is 0.296 versus 0.210 for the blocked certificate. It behaves like the plug-in rule because its effective sample size is large, so it selects the most aggressive empirical threshold. Third, budget utilisation is the main engineering

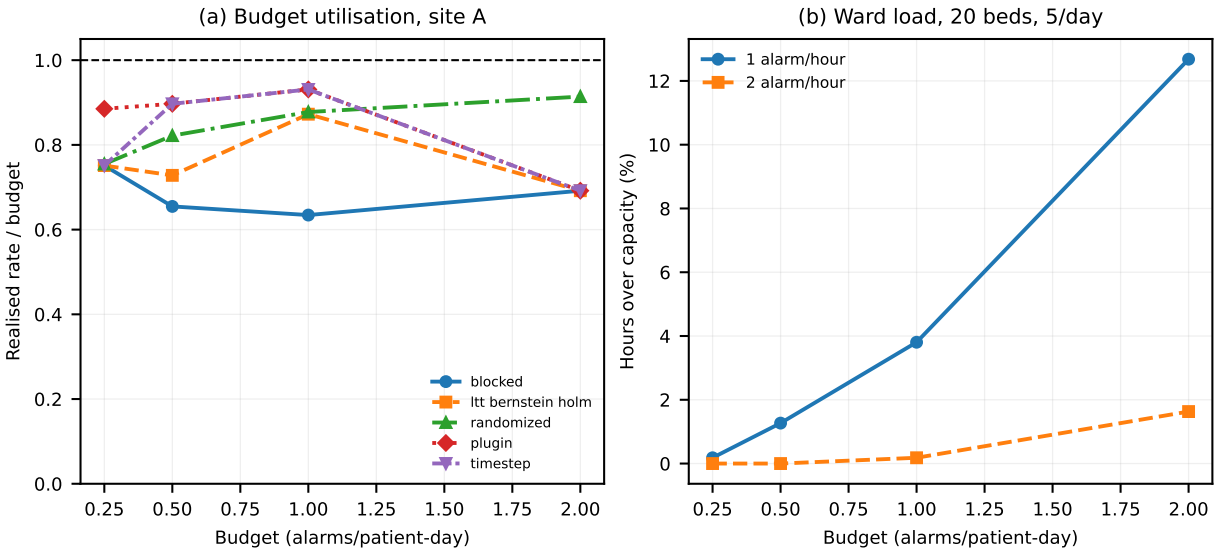


Figure 4: Operational summary. (a) Realised mean alarm rate divided by the requested budget over ten repeated patient splits on site A; the dashed line is exact utilisation. (b) Percentage of hours in which a simulated 20-bed ward with five admissions per day exceeds its alarm-handling capacity, using the calibrated traces from site A.

Table 9: Repeated-split coverage at the 1-per-patient-day budget, ten stratified splits per site. "Rate" is the mean realised alarms per patient-day, "Util." is rate divided by 1.0, "Viol." is the fraction of splits whose mean rate exceeded the budget, and "Patient viol." is the mean fraction of test patients above the per-patient budget. LTT is the empirical-Bernstein Learn-then-Test selector with Holm correction.

	Site A			Site B		
Procedure	Rate	Util.	Patient viol.	Rate	Util.	Patient viol.
Blocked (EB)	0.634	0.63	0.210	0.630	0.63	0.215
Timestep	0.931	0.93	0.296	0.935	0.93	0.308
Plugin	0.931	0.93	0.296	0.935	0.93	0.308
LTT (EB, Holm)	0.872	0.87	0.279	0.935	0.93	0.308
Randomised mixture	0.878	0.88	0.284	0.896	0.90	0.302

cost of the certificate. The blocked selector uses 63–75% of the budget on site A and 63–88% on site B across the four budgets. The empirical-Bernstein LTT and the randomised mixture raise utilisation to 69–91% without a mean violation in these splits, at the price of a higher patient-level violation fraction. The plug-in rule reaches 89–93% at the three tighter budgets (69% at 2 per day, where the grid binds) but has no finite-sample guarantee. This is the trade that the original single-split table could not show: the certificate is conservative, and the conservatism can be reduced by a sharper selector or by randomisation, not by pretending that the point estimate is a certificate.

5.8. Operational Alarm Metrics

Sensitivity and mean rate do not specify how many alarms a clinician sees for each detected event. We therefore report positive predictive value (PPV), the number needed to alert (NNA), and alarms per detected event, all computed from the same replay as the policy outcomes. On the primary A split at 1 per day, the blocked logistic policy emits 0.631 alarms per patient-day with PPV 0.060 and NNA 16.7; the plug-in policy emits 0.905 alarms per day with PPV 0.050 and NNA 20.0. On the external B evaluation the corresponding blocked values are PPV 0.037 and NNA 27.2, and the plug-in values are PPV 0.032 and NNA 31.4. At the tighter 0.25-per-day budget the blocked policy has PPV 0.099 and NNA 10.1 at A, because the same number of true alarms is divided by far fewer false alarms. The absolute PPV is low because the event prevalence is low and a patient can generate many hourly alarms; the useful engineering statement is the monotone trade: a tighter budget raises PPV and lowers NNA, while a looser budget buys sensitivity

Table 10: Target adaptation on site B, seed 0. The target cohort is split into disjoint warm-up and test halves. "Source" applies the A-calibrated threshold; "rate-matched" uses no target labels and matches the requested rate on the warm-up half; "oracle" recalibrates on target labels. Sensitivity and rate are evaluated on the target test half.

	Source		Rate-matched		Oracle	
Budget	Sens.	Rate	Sens.	Rate	Sens.	Rate
0.25	0.194	0.094	0.322	0.237	0.261	0.171
0.50	0.261	0.171	0.414	0.356	0.414	0.356
1.00	0.414	0.356	0.620	0.950	0.620	0.950
2.00	0.620	0.950	0.809	1.792	0.809	1.792

with more false alarms per detected event. These metrics also explain why the same sensitivity can correspond to very different clinical workloads across sites.

5.9. Label-Free Target Adaptation

The source-threshold audit shows that a threshold trained on A is too quiet on B. A target site that can observe unlabelled traces can still match its requested rate without waiting for outcome labels: it chooses the most aggressive threshold whose empirical alarm rate on a warm-up half of the target subjects is at most the budget, then deploys that threshold on the held-out half. Table 10 compares this rate-matched rule with source transfer and with an oracle that uses target labels.

On the target test half the label-free rule recovers the oracle operating point at 0.5, 1 and 2 per day: its threshold coincides with the target-calibrated threshold on the fixed grid. At 1 per day it raises sensitivity on that half from 0.414 to 0.620 and the realised rate from 0.356 to 0.950, still below the request. The source threshold is quieter on this half than on the full B test set, where the external audit reports 0.315 sensitivity and a 0.276 rate, because the warm-up/test partition changes the eligible-event mix. At 0.25 per day it selects a more aggressive threshold than the certified oracle, raising sensitivity from 0.194 to 0.322 at a rate of 0.237 rather than the oracle's 0.261 at 0.171. This is a positive engineering result, but it must be read narrowly. The rule is transductive: it uses a warm-up sample from the deployment site, and it controls the alarm rate rather than the risk. If the target case mix changes the relationship between score and event risk, a rate-matched threshold can meet the budget while detecting fewer events. The oracle agreement here is a property of this cohort and this grid, not a general guarantee. The finite-sample certificate still requires labelled target calibration patients; rate matching is a deployment heuristic that can be audited with the operational metrics above.

5.10. Ward-Level Alarm Load

Per-patient rates are difficult to translate into staffing. We replay the calibrated traces on a synthetic 20-bed ward with Poisson arrivals, finite beds and a fixed hourly alarm-handling capacity. At five admissions per day and a capacity of one alarm per hour, the blocked policy exceeds capacity in 0.2% of hours at 0.25 alarms per patient-day, 3.8% at 1 per day and 12.7% at 2 per day. At a capacity of two alarms per hour the corresponding fractions are 0.0%, 0.2% and 1.6% (Figure 4(b)). The peak hourly demand rises from two to three alarms, and the 95th percentile from one to two. At ten admissions per day the ward reaches 84% bed occupancy and refuses 13.6% of admissions at the 2-per-day operating point, so the load estimate is conditional on the arrival and capacity assumptions rather than a property of the policy alone. The simulation is a translation device, not a validation: it shows that a budget difference of a factor of two is operationally visible at the nursing-station level, and that a ward with one alarm per hour of spare capacity is stressed even at 1 alarm per patient-day.

5.11. Ablations

One-at-a-time ablations on the primary A split identify which design choices are first-order. The refractory window is one: moving from $\tau = 2$ to $\tau = 12$ raises sensitivity from 0.518 to 0.726 but also raises the patient-violation fraction from 0.167 to 0.327 at the 1-per-day budget. The escalation tier is a strong de-alarming control: $\kappa = 1$ lowers the realised rate from 0.631 to 0.472 and sensitivity from 0.607 to 0.286. The lead window changes sensitivity modestly

(0.560 at 6 hours, 0.607 at 12, 0.631 at 24) without changing the threshold. The risk level is second-order on this split: $\alpha = 0.05, 0.1, 0.2$ selects the same threshold, with certified risk changing from 0.844 to 0.823. Grid resolution is first-order at the coarse end: 20 thresholds produce a rate of 0.082 and sensitivity 0.179, 50 and 100 thresholds produce about 0.44 and 0.51, and 200 thresholds recover 0.631 and 0.607. The minimum-stay restriction changes both eligibility and the selected threshold: at 6, 12 and 48 hours the sensitivities are 0.627, 0.655 and 0.609 at rates 0.832, 0.825 and 0.747. These results do not justify a universal tuning rule, but they show that the reported operating point depends on declared structural choices, which is why those choices are fixed before calibration and reported with the certificate.

6. Discussion

The central empirical finding is that a patient-blocked alarm budget is portable as a *constraint* but not as a *threshold*. Within site A, the calibrated policy moves smoothly from 0.19 sensitivity at 0.25 alarms per patient-day to 0.77 at 2 per day, and the realised rates remain below the requested budgets. When the same A-calibrated threshold is applied to site B, the realised rate falls to roughly half the A rate and sensitivity at 1 per day falls from 0.585 to 0.315. The budget itself is still respected in the sense that B's realised mean is below the requested budget, but the clinical yield is much lower. This distinction matters for engineering practice: a site that imports only a threshold has not imported the operating point that the source site validated.

The comparison with the timestep shortcut clarifies what patient blocking buys, but not in the way a single-split table suggests. In ten repeated splits the shortcut stayed below the mean budget on both sites and often selected the same threshold as the plug-in rule, because its effective sample size is large. The data therefore do not show that timestep calibration fails empirically on this cohort. What they show is that the shortcut has no patient-level guarantee and that its patient-violation fraction is higher at 1 per day (0.296 versus 0.210). When the estimand is a patient-average rate, the effective sample size is the number of independent patients; a method that substitutes correlated hours can be less conservative without acquiring the guarantee it appears to have. The defensible conclusion is about the estimand and the audit trail, not a universal empirical ranking.

The backbone comparison is more nuanced than a single ranking. LightGBM has the highest site-B AUROC (0.713 ± 0.003) and the highest external sensitivity at 1 per day (0.540 ± 0.071), while the GRU has the highest site-A sensitivity (0.684 ± 0.006) and LightGBM is second (0.661 ± 0.042). The seed-0 paired LightGBM-minus-logistic sensitivity difference at B is 0.081 with interval $[0.027, 0.133]$ and Holm-adjusted $p = 0.066$, whereas the GRU-minus-logistic difference is -0.056 with an interval that includes zero after Holm adjustment. A stronger tabular scorer therefore narrows the external gap, but it does not remove the transport problem: LightGBM's B rate is 0.410 ± 0.156, still below the requested 1 per day, and the target label-budget analysis shows that local labels remain necessary to move the operating point. This is consistent with the broader observation that clinical prediction performance is often more sensitive to cohort and workflow differences than to modest architecture changes [36, 37, 58]. It also argues against presenting a backbone-only AUROC table as evidence of deployability.

For engineering deployment, the results support four concrete practices. First, the minimum observed stay used in the support bound should be declared before looking at calibration or test outcomes, because it changes both eligibility and $L_{\max}$; the 24-hour restriction here removes about 22% of A records and 26% of B records. Second, the no-alarm fallback should remain an explicit, auditable state rather than being replaced by the largest finite threshold, since the latter can still alarm and has no certificate. Third, a target site that has local labels should recalibrate on its own patient blocks before adopting a source threshold, because the external rate and sensitivity can move in opposite directions; our oracle split shows that this recovers part of the sensitivity but not without a higher alarm load. A label-free alternative can match the requested rate on unlabelled target traces, a one-shot empirical quantile rule related to adaptive conformal inference [59], but it controls the alarm rate rather than the risk and therefore cannot replace a labelled certificate. Fourth, the threshold grid and the realised-rate report should be published together, since a grid floor can make high budgets look identical and hide the fact that the constraint is no longer binding. These practices are operational rather than architectural, and they are compatible with the alert-stewardship literature that treats interruptive alerts as a monitored resource [7, 8, 48].

The study has several limitations that bound its claims. It is retrospective and uses a single public challenge cohort with two sites; the subject-disjoint split uses the official one-file-per-subject structure, but the release cannot link separate readmissions, so the independent-subject assumption remains an assumption. The label is the challenge's

sepsis definition, not a adjudicated clinical deterioration event, and left-censored records are excluded rather than imputed. The score models are deliberately simple reference learners, LightGBM is not tuned, and the GRU is not a state-of-the-art architecture. The partial NEWS, SIRS and qSOFA baselines omit components that the 2019 release does not contain, so they are research comparators rather than certified clinical instruments. The label-free adaptation study is transductive and controls only the alarm rate; it is not a coverage guarantee. The target-site recalibration and label-budget analyses are oracle diagnostics that use target labels and are not deployment procedures. No subgroup analysis, observation-sparsification study or clinical outcome evaluation was performed. ECE is low in absolute terms but is computed on a low-prevalence timestep problem and should not be read as clinical calibration. The primary external evaluation has no B calibration set by design, so it measures transport failure rather than the best achievable B performance. The three-seed E2 comparison is complete for all three backbones, but the target-site diagnostic for the GRU is single-seed, the label-budget curve is conditional on one target split and one policy family, the coverage study uses ten splits per site and therefore has coarse Monte Carlo resolution, and the study does not estimate GPU-hours or alert-fatigue outcomes.

These limitations do not overturn the methodological result, but they change what should be claimed. The paper does not show that budgeted calibration improves patient outcomes, reduces mortality or is preferred by clinicians. It shows that a stateful policy can be calibrated against a patient-level alarm budget with a finite-sample bound, that the resulting operating curve is measurable on a public cohort, and that the same threshold does not transport unchanged across hospitals. That is a narrower contribution than a clinical-effectiveness claim, but it is a more honest one, and it identifies the target-calibration experiment that a deployment study would need to run next.

7. Conclusion

This paper treated early warning as a stateful alarm policy whose alert load is a declared design constraint rather than an emergent threshold artefact. We derived a patient-blocked finite-grid upper-confidence procedure for the population mean of patient-level alarms per day, separated it from a deliberately invalid timestep effective-sample-size shortcut, and implemented the complete path from causal scoring through policy replay to auditable operating points. On the public 2019 A/B cohort, the procedure produced a monotone internal budget-sensitivity frontier, with A sensitivity rising from 0.185 at 0.25 alarms per patient-day to 0.774 at 2 per day over three seeds. The same A-calibrated threshold did not transport to site B: at the 1-per-day budget, B sensitivity was 0.315 versus 0.585 at A, even though the realised B alarm rate remained below the requested budget. An oracle target-site recalibration raised B sensitivity to 0.497 but also raised the realised rate to 0.616, and a nested label-budget analysis showed that roughly 10% of the eligible target subjects (about 1,482 patients) recovered the full-pool threshold at 1 per day while 5% remained unstable across seeds. LightGBM raised B sensitivity at 1 per day to 0.540 and the GRU reached 0.375, but neither removed the cross-site shift; the timestep shortcut supplied no patient-level guarantee and had a higher patient-level violation fraction at 1 per day, even though its realised mean rate remained below budget. Repeated-split coverage showed no mean-rate violation for the certified selectors across ten splits per site, with the blocked certificate using 63–75% of the budget and certified sharper selectors using up to 91%; a label-free rate-matching rule recovered the target operating point without target labels, and a ward simulation translated the difference between 1 and 2 alarms per patient-day into a visible change in hours over capacity. These results support budgeted, patient-blocked calibration as an engineering instrument, but they do not establish clinical effectiveness, alert-fatigue reduction or deployment readiness; subgroup and missingness analyses, prospective outcome evaluation and a utility-aware grid remain necessary before any clinical claim.

Data and code availability

The raw PhysioNet/CinC Challenge 2019 records are publicly available from PhysioNet under its data use agreement and are not redistributed with this manuscript. The analysis code, subject-split audit, score-cache builders and experiment drivers are included in the accompanying repository. Aggregated result files, including the seed-level bootstrap outputs, are archived under `code/experiment_results/` and `code/experiment_results/round2/`; raw records and score caches are not redistributed and can be regenerated from the public records with the documented commands.

Ethics statement

This study used only the public, de-identified PhysioNet/CinC Challenge 2019 cohort. No new patient data were collected, and no intervention or prospective clinical evaluation was performed. The analysis is a retrospective methodological study and does not constitute a clinical validation or a deployment recommendation.

CRediT authorship contribution statement

The author(s) contributed to the conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, writing of the original draft, review and editing, and visualisation. All numerical results were produced by the documented code and checked against the archived result files.

Declaration of competing interest

The author(s) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical approval

This study used only the public, de-identified PhysioNet/CinC Challenge 2019 cohort. No new patient data were collected, and no intervention or prospective clinical evaluation was performed. The analysis is a retrospective methodological study and does not constitute a clinical validation or a deployment recommendation.

Use of generative artificial intelligence

Generative artificial-intelligence tools were used for language editing and code assistance. The author(s) verified all claims, references, numerical results and code, and take full responsibility for the content of the manuscript.

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